

ASSIGNMENT 1

DATA MINING

Submitted by

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KDD Process in Digital Library Recommendation Data Mining

In digital library recommendation systems, the KDD process is used to analyse large volumes of reading, borrowing, and search behaviour data in order to understand user interests and improve book/article recommendations.

Step-1: Data Pre-Processing

The digital library usage records are cleaned and prepared before analysis.

This includes:

- removal of irrelevant events (auto-refresh, repeated page loads)
- handling missing or incomplete user activity trails
- removal of duplicate transactions (same borrow logged multiple times)
- identification of unique users, sessions, and reading histories

Assumption and justification

Automated refresh actions and duplicate transactions do not represent genuine reading behaviour and may distort interest patterns. Hence, they are removed.

Step-2: Data Exploration

The cleaned library data is transformed into a suitable structure for mining.

This includes:

- grouping user actions into reading sessions
- creation of sequences of viewed/borrowed topics
- aggregation of total reads per subject category

Statistical summaries such as most accessed subjects and average session duration are computed.

Step-3: Data Mining

Data mining algorithms are applied to discover reading and interest patterns.

Common techniques:

- association rule mining to discover frequently co-read topics

- sequential pattern mining to analyse reading order behaviour
- clustering to group similar reader profiles

Step-4: Data Presentation

The discovered patterns are evaluated and visualized.

This includes:

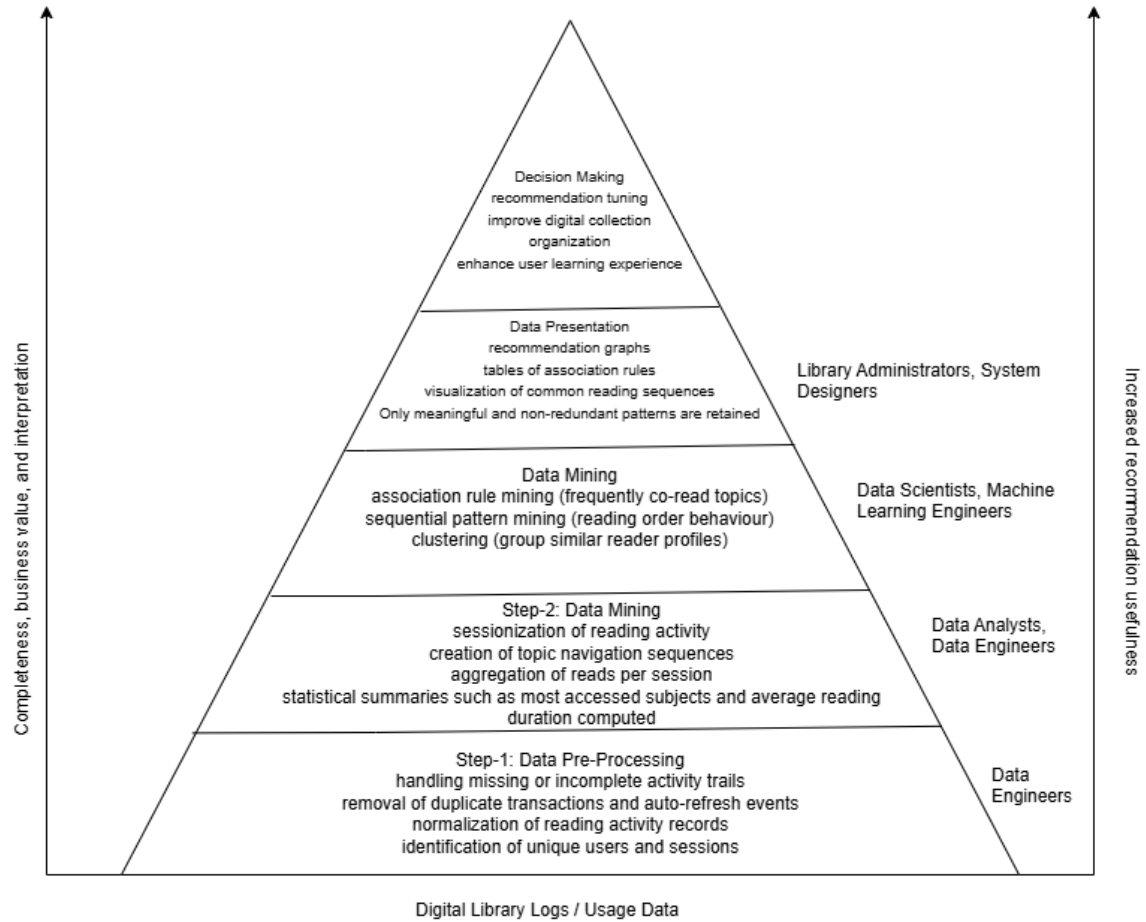
- recommendation graphs
- tables of association rules
- visualization of common reading sequences
- Only meaningful and non-redundant patterns are retained.

Step-5: Decision Making

The extracted knowledge is used by library administrators and system designers to:

- improve recommendation accuracy
- organize digital collections better
- enhance user learning experience

KDD Process in Digital Library Recommendation Data Mining



KDD Process in Traffic Accident Hotspot Data Mining:

In traffic accident hotspot mining, the KDD process is used to analyse accident reports and road-condition datasets in order to identify high-risk zones and factors contributing to accidents.

Step-1: Data Pre-Processing

Accident records are cleaned and prepared before analysis.

This includes:

- handling missing or incomplete accident reports
- removal of duplicate accident entries
- normalization of numerical attributes (speed, time, weather index)
- anonymization of sensitive driver information

Assumption and justification

Incomplete and duplicate accident reports may lead to misleading risk patterns, and driver privacy must be protected. Hence, data cleaning and anonymization are performed.

Step-2: Data Exploration

The cleaned accident data is transformed into a suitable format for mining.

This includes:

- aggregation of accidents by location, time, and road type
- creation of derived attributes (accident severity score, risk index)
- analysis of frequency of accidents across zones

Statistical summaries such as accident distribution and severity occurrence are computed.

Step-3: Data Mining

Data mining algorithms are applied to discover accident-related knowledge.

Common techniques:

- association and correlation analysis to identify relationships among road conditions and accident severity

- classification to predict accident severity categories
- clustering to group similar accident zones

Step-4: Data Presentation

The discovered patterns are evaluated and presented.

This includes:

- tables of association rules
- charts showing location-risk relationships
- visualization of accident hotspot clusters
- Only practically meaningful patterns are retained.

Step-5: Decision Making

The extracted knowledge supports transport authorities to:

- improve road safety planning
- prioritize hotspot monitoring
- support accident prevention strategies

KDD Process in Traffic Accident Hotspot Data Mining

